Workshop 2.3 Machine Learning vs. Deep Learning

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Introduction

In this assignment, I will explain the difference between Machine Learning and Deep Learning by providing a real-world application for each type of technology, explaining why the approach is suitable for the problem, and why the other approach is not suitable.

Machine Learning

Machine learning can be used for image recognition in autonomous, affordable home robotics, sensors, or smart appliances. Some specific products are vacuum robots, lawn trimming robots, smart doorbells, human/pet presence sensors, refrigerators (for tracking expiration dates of food), and interactive multimedia smart assistant devices.

Machine learning is a better fit for this application because it consumes much less energy and computational resources than deep learning. In a hardware product, choosing machine learning means a less powerful CPU and a smaller battery can be selected. For products that usually sells for under $100, this means significant savings in part cost and a higher profit margin. Although deep learning has better capabilities, the economy does not allow it in affordable products.

Even when the task is complex enough to justify having deep learning models running on expensive hardware, traditional machine learning can still be utilized as a backup solution. In fact, most devices stay idle most of the time, and wake up only when actively being used or events of importance occur. A machine learning algorithm can run during sleep and consume very little energy. When it detects events, it wakes up the whole system and the deep learning capabilities to better handle the situation. Some examples of this behavior can be a smart doorbell seeing someone passing by the front door, or a robot vacuum only running deep learning when it detects significant staining on the floor. By using machine learning for the majority of the time, products can improve energy efficiency and extend battery runtime.

Deep Learning

One potential application for deep learning is the on-the-fly generation of voice and speech for characters in video games. Many game genres can benefit from this solution, for example, RPG, sports, action, and adventure.

Deep learning is a suitable solution in this field is because it solves many problems and does not have much constraints. The first problem it solves is the redundancy of audio files in the game. Currently, each line of text must be per-recorded, and frequent speeches need to have

multiple variants to feel more natural when repeated. By using deep learning, tens or even hundreds of gigabytes of files can be eliminated. The second problem is situational fitness of the text. For sports games, where a commentator is present to mimic the feel of live TV, audio recordings might struggle to describe the actual situation of the game. With deep learning, fixed lines aren't necessary, as the model can describe the game without picking from per-written sentences. It can also generate voices for different languages where a voice actor with a similar voice does not exist, improving consistency of game experience across regions. As games already run on more powerful computer systems, the amount of processing is less of a constraint compared to micro-controller-based products.

Voice synthesis is possible with non-deep-learning approach. However, in video games, where the context can be highly dynamic, it falls short on creating emotional tones in the voices and making the characters sound natural. The last thing a player want is to hear a "robot voice" when they are supposed to be talking to a (virtual) human. Using traditional machine learning for this application is not a good approach as it significantly degrades the player's experience, and breaks their immersion from the world of the game.